

HW04 Main Report — Automation Testing

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Course: CS423 / CSC13003 – Software Testing

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Package: SoftwareTesting-HW/HW4/23127271/ · GitHub: trngnneee/eshop-sut branch HW4-Tram

1. Scope

Pool	Feature	Spec cases	Spec file	Data file
A	FR-03 Forgot Password (Web)	14	tests/fr03-forgot-password.spec.js	test-data/fr03-forgot-password.json
B	FR-08 Checkout	14	tests/fr08-checkout.spec.js	test-data/fr08-checkout.json
C	FR-15 Product CRUD (Admin)	14	tests/fr15-admin-product.spec.js	test-data/fr15-admin-product.json

HW04 Task 1 requirements addressed: data-driven Playwright, ≥ 12 cases/feature, ≥ 3 assertion patterns, Chromium/Firefox/WebKit, HTML reports labeled Run by: 23127271 .

2. AI collaboration (step-by-step)

Each feature followed: **Analyze** → **Design** → **Review** → **Model data** → **Map** → **Generate** → **Verify**.

Evidence:

- Stage log: docs/ai-conversion-log.md
- Prompt appendix: docs/prompt_log.md
- AI Audit (A+B+C): docs/ai-audit-report.md (+ PDF)
- AI Critique (200–300 words): docs/ai-critique.md (+ PDF)

Human review kept **spec oracles** when the SUT was defective (OTP length, checkout auth/total, product validation / FR-12).

3. Assertion patterns (≥ 3)

Pattern	Example API	Used in
Visibility / hidden	toBeVisible / toBeHidden	FR-03, FR-08, FR-15
Text / accessible name	toHaveText / toContainText	FR-03, FR-08, FR-15
URL / navigation	toHaveURL	FR-03, FR-08
Value / attribute / count	toHaveValue / toHaveAttribute / toHaveCount	FR-03, FR-08
Plain / API status	expect(status).toBe(...) ranges	FR-08 totals, FR-15 API

4. Multi-browser execution summary

Feature	Chromium	Firefox	WebKit	HTML path
FR-03	9 pass / 5 fail	9 / 5	9 / 5	reports/html/fr03-forgot-password/<browser>/
FR-08	9 / 5	9 / 5	9 / 5	reports/html/fr08-checkout/<browser>/
FR-15	6 / 8	6 / 8	6 / 8	reports/html/fr15-admin-product/<browser>/

All nine cells stamp **Run by: 23127271** (title + header). Manifest: `reports/run-manifest.json` .
Feature A reports are **frozen** (`EVIDENCE-LOCK.json` + `evidence/feature-a-fr03-frozen-2026-08-07/`).
Failing cases are **product defects**, not softened oracles. See `bug-reports/` .

5. Bug reports

Feature	Local Markdown	Count
FR-03	bug-reports/BUG-FR03-001 ... 005	5
FR-08	bug-reports/BUG-FR08-001 ... 005	5
FR-15	bug-reports/BUG-FR15-001 ... 008	8

GitHub Issues filed on `trngnneee/eshop-sut` with screenshots (links updated in `bug-reports/README.md`).

6. Agent Skill

Item	Detail
Skill file	<code>.cursor/skills/automation-testing/playwright-skill.md</code>
Skill demo (FR-15 E2E)	https://youtu.be/Te25xh0biYI
Task 2 demo (FR-15 run + 3 browsers + HTML + AI fix)	https://youtu.be/p5TRjyrvPvg

7. How to reproduce

```
cd SoftwareTesting-HW\HW4\23127271
# API :3000, Web :5173, Admin :5174
npm run evidence:verify-fr03
npm run test:matrix:fr08
npm run test:matrix:fr15
```

GitHub automation mirror: `Repo/eshop-sut/hw04-automation` on branch `HW4-Tram` .
Commit history for `.spec.js` : `git-commit-log.txt` .

8. Self-assessment

No.	Criteria	Max	Self
1	Feature A (FR-03)	25	22
1	Feature B (FR-08)	25	22
1	Feature C (FR-15)	25	22
2	Task 2 — Demo video	15	15
3	Agent Skills	10	9–10
	Total	100	~90–91

9. AI gap review (brief)

AI accelerated scaffolding but repeatedly risked green-washing defective SUT behavior. Mandatory human gate: keep README/FR oracles, fix only automation bugs (dialogs, SPA cart seed, locators), and treat matrix fails as either product bugs or environment blockers — never as a reason to edit expected results.